



TRANSFORMING
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UNIVERSAL USB POWER SUPPLY WITH INTELLIGENT SUPPLY VOLTAGE REGULATIONS APPLICATION

具備智能供電電壓調節的通用USB電源

Nowadays, recharging mobile gadgets have become parts of our daily life. USB power supply is becoming an indispensable accessory for many gadgets, such as smartphone, tablet and digital camera. Yet, other mobile gadgets such as laptop computer are always bundled with a dedicated power supply unit. Many users end-up carrying several power supplies in their bags.

Targeting to relieve the carrying burden of users, this intelligent USB power supply not only recharges smartphones, tablets and digital cameras, but also laptop computers through an ordinary USB outlet. By using an intelligent detection and control algorithm, the voltage output of the intelligent power supply is regulated automatically to meet the voltage rating of the mobile gadgets. The firmware of this intelligent USB power supply can also be updated through the USB outlet so that power supply requirements for new mobile gadgets can be accommodated. The USB power supply can also provide power to passive loads such as mobile fans and portable lightings.



現今，手機充電器已經成為了我們日常生活的一部分。USB電源也成為了像智能手機、平板電腦、數碼相機等的充電設備中不可或缺的部分。然而，像筆記本電腦之類的移動設備仍舊需要專用的充電/供電設備，這使得用戶還是需要在他們的包裡攜帶多個電源。

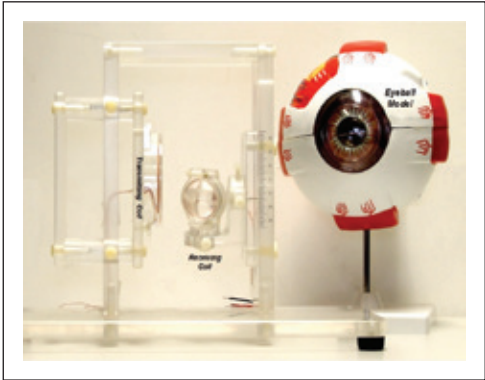
為了減輕攜帶負擔，本款智能USB電源不僅可以給智能手機、平板電腦和數碼相機充電，而且也可以通過普通的USB插口作為筆記本電腦的電源。通過一個智能檢測與控制算法，該智能電源能調節其輸出電壓以滿足移動設備的額定電壓。其固件可以通過USB口進行更新，以適應新的移動設備的電源要求。它還可以為類似移動風扇和便攜式照明設備等被動負載提供動力。

MISALIGNMENT TOLERABLE

COIL STRUCTURE FOR BIOMEDICAL IMPLANTS

生物醫學植入裝置適用之容許位置偏差線圈結構

Coil-misalignment is one of the major hurdles for inductively coupled wireless power transfer in applications such as retinal prosthesis. Weak magnetic flux linkage due to coil misalignments would significantly impair the power efficiency. A novel receiver configuration with high misalignment tolerance is presented. The proposed receiver is composed of two receiver coils placed orthogonally, so as to reduce the variation of mutual inductance between transmitting and receiving coils under misalignment conditions. These two receiving coils can be in the form of series coil connection (SCC) or parallel coil connection (PCC). The proposed structure offers an omnidirectional coupling. Two coils are simultaneously energized to guarantee to operate at the achievable maximum power transfer.

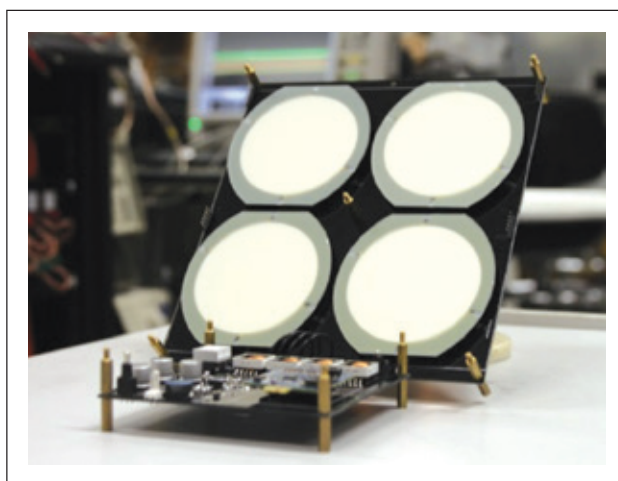


線圈偏差是電感耦合無線電力傳輸應用中的一個主要障礙。如在人工視網膜裝置應用中，由於線圈移位，弱磁通鏈會顯著削弱傳遞效率。新型線圈結構可容許大幅度的線圈偏差，新型接收器是由兩個垂直放置的接收線圈組成，以減小因線圈偏差引致發射線圈和接收線圈之間的互感系數變化，而這兩個接收線圈可以串聯或並聯形式連接。新型線圈結構提供了一個全方位的電感耦合，在運作期間，兩個線圈均同時通電，以達致最大可能的功率傳輸。

INDUCTIVE POWER TRANSFER SYSTEM FOR DRIVING MULTIPLE OLED/LED LIGHTING PANELS

用於驅動多個OLED/LED的無線輸電照明系統

This project is an inductive power transfer system for driving multiple organic light-emitting diode (OLED) / LED lamps. Instead of using electricity, the magnetic field is induced by a detachable transformer to transfer energy to each lamp so that a safely hot swappable and waterproof condition is achieved. All the lamps are in paralleled connection and remained in the same brightness by our patented Daisy-chained transformer (DCT) technology. The whole system is powered by only one switching network to realize simple control, low cost and low loss. Meanwhile, an intelligent sensing system is embedded to detect whether the lamps is well aligned to the lamp holder and bypass the broken lamps, which also reduces the loss and improve the system's stability.

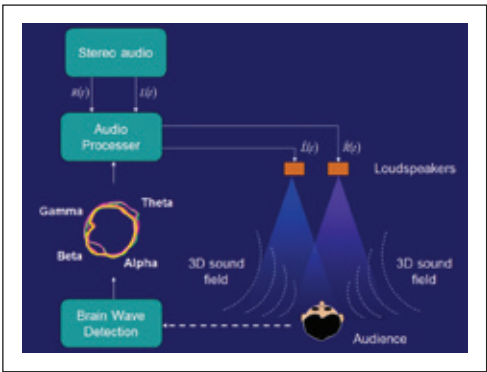


本項目主要用於驅動多個LED/OLED照明燈具，每個燈具與燈座之間通過可拆分變壓器連接並利用磁場傳輸能量，避免了帶電熱拔插燈具時造成的電火花，從而實現了燈具的安全熱拔插和防水功能，燈具之間為並聯連接並通過我們已獲得專利的鍊式變壓器技術來保持不同燈具之間的亮度一致性，整個電路僅由一個開關電源模塊驅動，因而控制簡單、成本和損耗較低。同時我們還加入了智能化的感應系統來檢測燈具是否安裝到位以及頻閉損壞燈具，從而進一步減小了損耗並提升了系統的可靠性。

NEURAL INDUCED ENHANCEMENT OF AUDIO SIGNALS FOR PROMOTING LISTENING PLEASURE

神經誘導增強音響訊號以提昇聽覺享受

The product is designed based on a method with which listeners can exert control (consciously or sub-consciously) over various audio enhancement effects through their mental states. Briefly, brain waves at different frequency bands are detected from a listener, and the signal in each frequency band is employed to adjust one or more audio enhancement effect(s). A listener can choose to perform this process subconsciously, in which case the audio enhancement merely follows the mental state or emotion. On the other hand, the listener can also modify one's brain waves intentionally by invoking certain mental activities (such as deliberately increasing, or decreasing the attention).

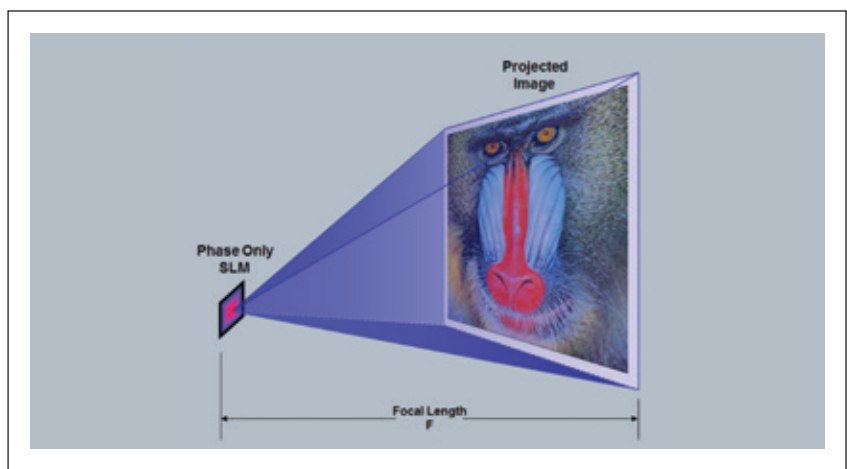


本產品的基本設計方法為聽眾可以透過他們的精神狀態控制（有意識或潛意識地）不同的音響加強效果。簡單來說，要檢測聽眾不同頻帶的腦電波，而每個頻帶的訊號將同時被用來調節一個或更多的音響加強效果。聽眾可以選擇在潛意識下進行這過程，此時音響加強效果僅僅跟隨聽眾的精神狀態或情緒改變。另一方面，聽眾也可以通過借助某些心理活動（如故意增加或減少注意力），有意地改變其腦電波。

LOW COST HOLOGRAPHIC PROJECTOR 低成本全息投影儀

Recently, there is an increasing trend of decreasing the size, as well as enhancing the quality of optical projectors. An approach to achieve this goal is through the use of "holographic projection". In brief, the hologram of an image is projected with an LCOS (Liquid Crystal on Silicon) device instead of the actual image. There is no need to incorporate the bulky optical lens for focusing, hence allowing the system to be implemented in small size and with lower cost. In addition, the projected image is insensitive to dead pixels in the display devices.

This invention describes a method for generating a "constant magnitude hologram" that can be represented simply by its phase information. As such, the hologram can be taken as a "pure phase hologram" that can be displayed with a phase only spatial light modulator (SLM) with good efficiency. The reconstructed image is free from the zero order diffraction and the twin image. Different from the existing methods, no further processing is required once the hologram is generated. The recorded 3D image can be reconstructed simply by illuminating the pure phase SLM with a coherent beam. The speed in generating the hologram is as fast as the one used to generate a digital Fresnel hologram, which with the present state of the art, can be accomplished at over 40 frames per second for a medium size hologram comprising of over 4 million object points. The quality of the reconstructed image is superior to existing fast techniques.



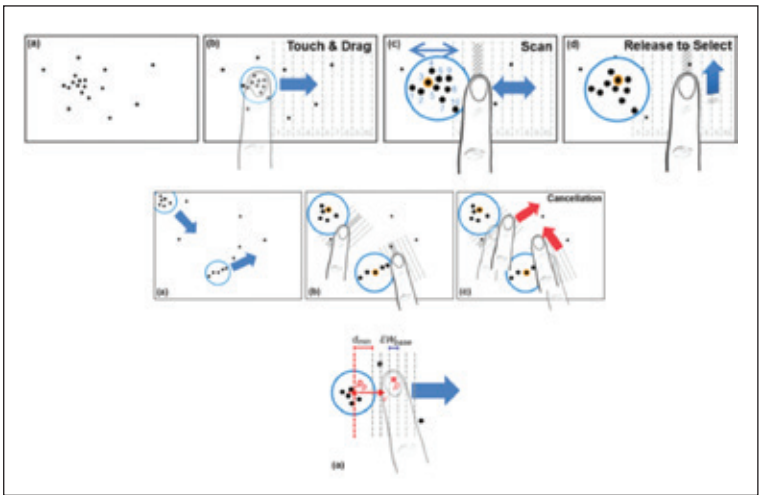
最近，光學投影儀有尺寸變小及質量提高的趨勢。實現這一目標可採用一種「全息投影」的方法。簡單來說，使用LCOS設備可投影出圖像的全息圖，而非實際圖像。因此，無需整合笨重的光學透鏡來實現聚焦，使該系統實現小尺寸和更低成本的運行。此外，顯示裝置的壞點對投影圖像不會造成影響。

本發明描述了一種生成「恒定幅度全息圖」的方法。這種全息圖能簡單地用其相位信息來表示。因此，全息圖可被視為一個「純相位全息圖」，通過純位相空間光調製器（SLM）有效地顯示出來。重構後的圖像消除了零級衍射及共軛像。與目前方法不同，該全息圖生成後無需進一步的處理。所記錄的3D圖像可簡單地通過相干光束照射純位相SLM進行重建。生成全息圖的速度與用於生成數字菲涅爾全息圖的速度相當。採用目前技術發展水平，可在每秒40幀情況下生成一個含400萬個對象點的中等大小的全息圖。重構圖像的質量要優於現有的快速技術。

LINEARDRAGGER: A LINEAR SELECTOR FOR ONE-FINGER TARGET ACQUISITION

線性拖拽：於觸控界面使用的單指目標獲取界面

This project introduces LinearDragger, a new one-finger target acquisition technique for small and clustered targets. Our key contribution is a mapping approach to convert the 2D targeting selection problem to a 1D selection problem, with a simple touch-drag-release operation. Our technique gives constant effective width for all candidate targets, which is independent of the target distribution. This leads to a predictable and controllable selection interface. Results of our controlled experiments show that LinearDragger is a promising target acquisition technique for touch devices, and performs better than existing popular selection techniques that involve a single-touch operation, in terms of both selection time and error rate.

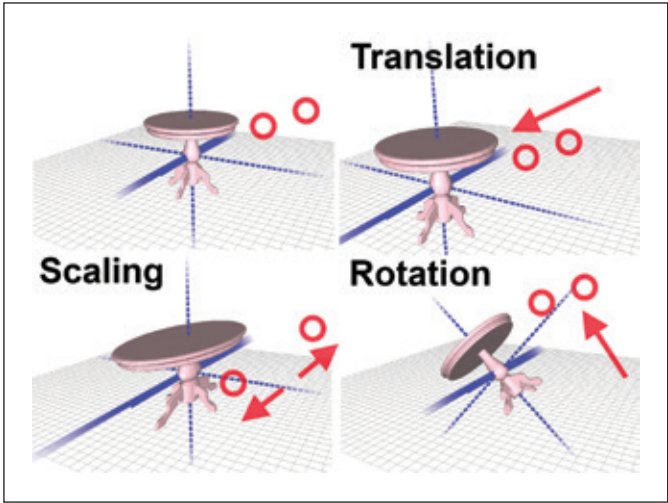


本項目介紹了線性拖拽，一種用於小且群集的目標的新型單指目標獲取技術。本發明一個新的貢獻是利用簡單的觸摸-拖拽-釋放操作將2D目標選擇問題轉換成1D選擇問題的一種映射途徑。這一新穎技術給予了所有候選目標恆定的有效寬度，其不依賴於目標分佈。這得到了可預測並可控制的選擇接口。我們的控制實驗的結果顯示線性拖拽是一種有前途的用於觸摸裝置的目標獲取技術，並且在選擇時間和誤差率方面，線性拖拽比涉及單個觸摸操作的現存的流行選擇技術表現得要好。

MULTITOUCH GESTURES FOR CONSTRAINED TRANSFORMATION OF 3D OBJECTS

用於三維物件約束操控的多點觸控界面

This project introduces a widgetless and buttonless multi-touch interface for constrained manipulations of 3D objects. The widget-less multi-touch user interface can map different multi-touch gestures to different 3D object manipulation tasks, allowing a user to manipulate 3D objects without operating widgets or performing mode switchings or tool selections. User interaction is greatly simplified by using single touch actions that simultaneously specify the transformation constraint, the transformation type, and the magnitude of transformation and apply the transformation to the focus object. Additionally, the axis-based constrained manipulations support active snapping and axis transfer. Active snapping allows the user to draw a free touch path connecting two 3D objects to be snapped together, avoiding tedious operations required with standard transformation manipulations. Axis transfer provides a simple solution to achieve relative transformations between objects, allowing an object to be transformed along a specific predefined axis of another object.



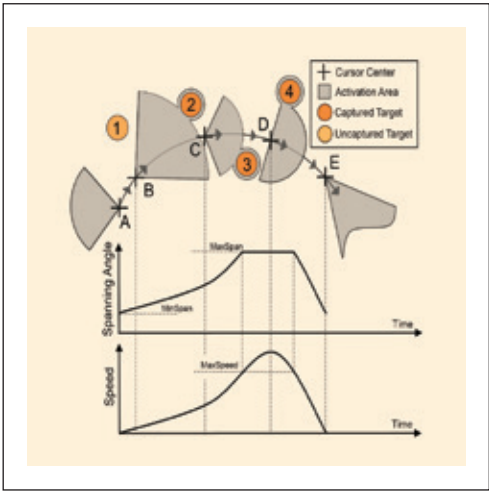
本項目所述的是採用了一種無小部件及無按鈕的多點觸控界面，用於三維物件的約束操控的系統和方法。無小部件的多點觸控用戶界面可以將不同的多點觸控手勢映射到不同的三維物件操控，允許使用者直接操控三維物件，而不需要操作小部件、執行模式切換或選擇不同的工具。通過同時指定變換約束、變換類型和變換的大小，以套用變換至專注中的物件，使用者互動可被大大簡化。此外，以軸線為基礎的約束操作支持主動貼緊和軸線轉移。主動貼緊允許使用者以連接兩個三維物體的自由觸控路徑來貼齊此兩個三維物體，避免了標準變換操作所需的繁瑣操作。而軸線轉移提供了一個簡單的解決方案來實現物件之間的相對變換操作，使一物件可沿另一特定物件預先定義的軸線作變換。

THE IMPLICIT FAN CURSOR: A VELOCITY DEPENDENT AREA CURSOR

隱式扇形游標：基於速率的區域游標

We present the Implicit Fan Cursor (IFC) – a novel target pointing technique using a cursor with a fan-shape activation area. The IFC couples the cursor’s activation area with its velocity, i.e., the speed and direction of the mouse motion, behaving like a 2D spotlight cursor at low speed and a circular area cursor at high speed.

IFC enables the user to precisely acquire distant targets at low speed and easily acquire the nearest targets at high speed, without explicit mode switching. This technique minimizes cursor movements, while taking into consideration the precision of cursor movements at different speeds. It also ensures that only one target is captured at any time. The results of our controlled experiments show that the IFC outperforms the point cursor and the area cursor techniques, particularly in terms of cursor moving distance.



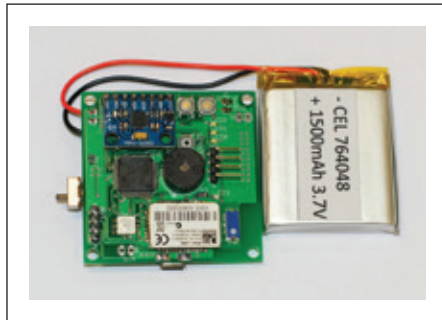
我們在此介紹隱式扇型游標 (IFC)：一個基於扇形激活區域的新型目標選取技術。游標的速度和方向定義了激活區域的大小及形狀，使其在低速時如一個2D聚光燈光，及在高速時為圓形的區域游標。

使用者在游標低速時能夠精確地選取遠處的目標，及於高速時輕易地選取接近的目標，而不需要任何模式切換。此技術減少了游標操控及滑鼠移動，同時考慮了游標在不同速度時的精確度需求。實驗的結果表明IFC優於傳統游標和其他區域游標，尤其在於游標移動距離而言。

PORTABLE MOTION DETECTION AND RECOGNITION DEVICES

便攜式移動偵測及識別裝置

A motion detection device is developed in this project. The device can measure and record human motions. It can also recognize motion patterns, such as falling, jumping, etc. The device can send data to computers, cloud storages, mobile phones or alarms through various wireless communication channels. It is portable and can even be embedded in another device, such as a smart watch.

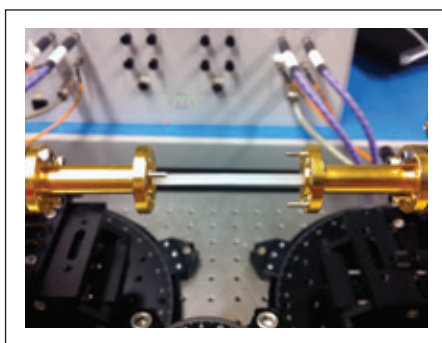


這個項目開發了一個運動檢測裝置。該裝置可以測量和記錄人的動作，也可以識別運動模式，如摔倒、跳躍等。它可以通過各種無線通訊信道將數據發送到電腦、雲存儲、移動電話、警報等等。便攜式的裝置，亦適合被嵌入到如智能手錶等另一個設備上。

AN INTEGRATED TERAHERTZ ANALOG CIRCUIT (ITAC) PLATFORM

太赫兹集成波導平台

A low-cost process is developed for the mass production of low-loss dielectric waveguide circuits for application in the sub-millimeter and terahertz in frequency range between 50 GHz and 1 THz. The proposed process utilizes conventional injection molding technology to produce functional waveguide circuits that comprises single or multiple materials and can be connected directly to a metallic waveguide interface of the submillimeter wave components. The samples can show the potential of the proposed platform to be used in the development and manufacturing of basic dielectric waveguide components in submillimeter wave and terahertz applications.

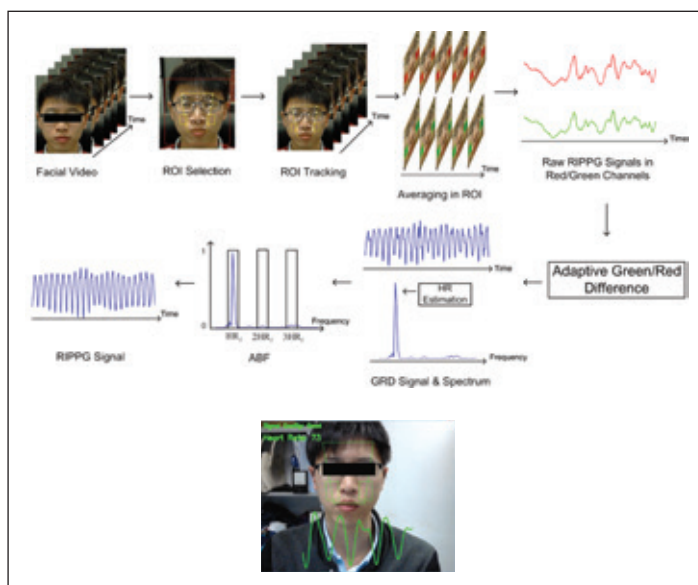


此低成本工藝可用於批量生產低損耗介質波導電路，應用於50GHz至1THz頻率範圍內的亞毫米和太赫茲波段。此工藝使用傳統的注模技術生產功能性波導電路，該電路包含一種或多種材料，並能夠與亞毫米波元件的金屬波導介面直接相連。已製成的樣本證明ITAC平台具備開發和生產應用於亞毫米和太赫茲波段基本介質波導元件的潛力。

REMOTE PPG BASED HEART RATE MONITORING FROM HUMAN FACE

基於非接觸式PPG的人臉心跳監測

Remote photoplethysmography (rPPG) can achieve contactless heart rate monitoring using a digital camera. Since hemoglobin in blood can absorb light, the color of human skin varies slightly with blood pulses. Though this color variation of the skin due to blood pulse is invisible to human eyes, it can be detected by a digital camera using signal/video processing techniques. We proposed a robust heart rate estimation algorithm based on rPPG, which can implement real-time remote heart rate monitoring from human face using only a webcam with ambient light. With the aid of face tracking and pulse activity detection, the remote heart rate monitoring algorithm can achieve an acceptable measurement accuracy.

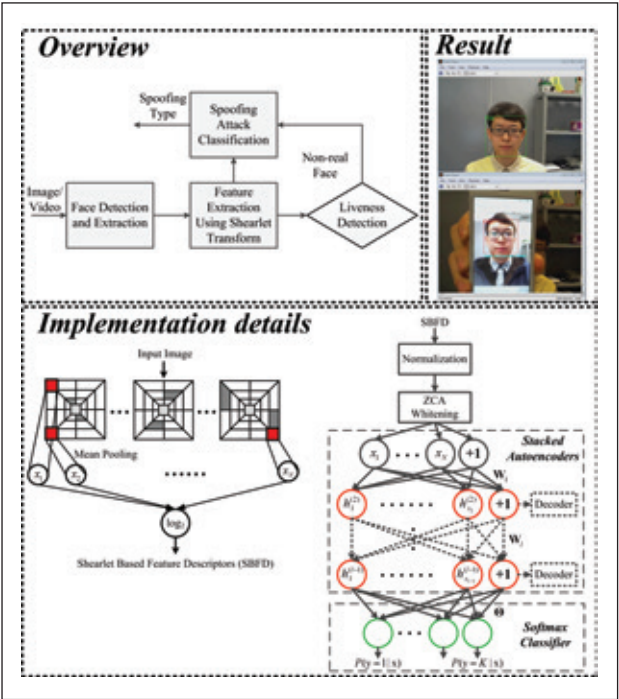


非接觸式PPG技術能夠利用數字攝像頭實現非接觸式心跳監測。人體血液裏的血紅蛋白具有吸光特徵，人體皮膚的顏色會隨著脈搏跳動有微弱的變化。這種脈搏引起的人體皮膚顏色變化人眼不可見，但是利用信號和視頻處理技術，可以被數字攝像頭檢測出來。我們提出了一種基於非接觸式PPG的穩健的心跳監測算法。該算法可以在日常光線條件下僅使用網絡攝像頭就可以從人臉讀出實時心跳。人臉跟蹤和脈搏活動檢測技術使得該非接觸式心跳監測算法具有可靠精度。

SHEARLET BASED FACE LIVENESS DETECTION FOR FACE AUTHENTICATION APPLICATIONS

基於剪切波變換的人臉活性檢測

Face recognition is a widely used biometric technology due to its convenience but it is vulnerable to spoofing attacks made by non-real faces such as photographs or videos of valid user. Anti-spoof problem must be well resolved before widely applying face recognition in our daily life. In which, face liveness detection is a core technology to make sure that the input face is a live person. The aim of this project is to propose a feature descriptor and an efficient framework which can be used to effectively deal with face liveness detection problem. In this framework, new feature descriptors are defined using a multiscale directional transform (shearlet transform). Then, stacked autoencoders and softmax classifier are concatenated to detect face liveness.

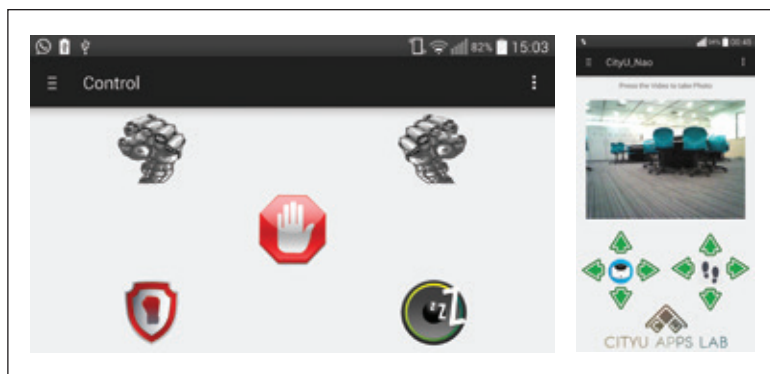


人臉識別因其方便性已經成為一種廣泛使用的生物識別技術。然而，這種技術很容易遭受欺騙攻擊，例如真實用戶的照片和視頻。因此，在將人臉識別技術廣泛應用於日常生活之前，必須有效的解決欺騙攻擊問題。人臉活性檢測就是其中一項核心的技術，該技術確保輸入是一個真實的人臉。該項目的目的正是提出一種特徵和框架來有效地處理人臉活性檢測問題。在該框架中，我們利用多尺度方向變換（剪切波變換）提取圖像特徵。然後將多層自編碼器和softmax分類器連接用來對特徵進行分類，從而檢測人臉活性。

ROBOTDROID

機械人遙控器

“RobotDroid”, meaning controlling robot via Android, can be used as an ideal tool to control the motion of NAO Robots through mobile phone. A very positive response has been elicited during last year’s ICT Expo by the nice features our engineering design, which include instruction-based robotic movement and punch throwing, and voice-command-based dictation. This year, the functionality of our app is further upgraded, covering new features including 1) 3D inspection of surroundings utilizing real-time video stream captured by vision sensors on the robots; 2) new human-tracking mode allowing the robots to track and “stare” at any persons nearby; 3) novel memory ability allowing the robots to perform any predefined motion after a one-off “teaching” session.



此「機械人遙控器」是一款通過手機終端來實現對機械人行動操控的數位工具。該流動程式在去年的國際資訊科技博覽會一經展出，即引起熱烈迴響。當時此流動程式之基本功能包括對機械人姿態/動作（包括空間移位，揮拳擊打等）之控制和語音命令之傳輸等。今年，該流動程式經過進一步更新升級，其功能亦更為強大和全面。這些新的功能主要包括：1）基於圖像傳感器和實時視頻流獲取的三維空間環境掃描；2）允許機器人檢測附近人物之全新人體跟蹤模式；3）允許機械人執行任何預錄動作之全新學習/記憶模式。

SMART BASE STATION ANTENNAS FOR NEW GENERATION BROADBAND MOBILE COMMUNICATIONS

新一代寬帶移動通訊的智能基站天線系統

While standardization of the Fourth Generation (4G), also known as Beyond 3G, is still undergoing deliberation, this all-IP system will certainly surpass the quality of previous generations in delivering voice, data and streamed multimedia to users, allowing seamless and transparent communications.

The interoperability of the 4G system requires wideband and high power antenna system operations. On the other hand, access techniques to heterogeneous networks require more efficient use of energy. Intelligent antenna is another essential piece to complete the 4G jigsaw puzzle in providing high speed and highly secured communications.

In this project, we draw upon the expertise and facilities of the State Key Laboratory of Millimeter Waves at City University of Hong Kong, Comba Telecom System (Guangzhou) Limited, the number one integrated wireless solutions and subsystem provider in Asia, and Jackson Manufactory Dongguan Limited, a manufacturer for mobile communications products, to collaborate on the endeavor of developing an enabling technology in wideband adaptive radio frequency (RF) subsystem for the 4G mobile communications.

Specifically, we will develop wideband antenna elements, high linearization technology for power amplifiers, and novel up- and down-conversion technologies with phased sources. The latter two will be implemented in integrated circuits. All the technologies will be employed to design adaptive antenna arrays for the 4G mobile communications.



4G（常稱後3G）的標準化正進行審慎的研究。這個全-IP系統透過無線、透明的通訊管道傳送聲音、數據及多媒體給予用戶。相對於現有的通訊技術，4G系統在素質上必更為優勝。4G系統需寬頻或多頻來達致軟硬體間的良好互通性。此外，在存取不同網絡時，我們需要更有效的能源使用，所以，在4G上實行高速而安全的通訊，智能天線是不可或缺的。在這項目中，我們集合毫米波國家重點實驗室（香港城市大學）、京信通信技術（廣州）有限公司（全球最大的集成無線解決方案及子系統供應商之一）及東莞積信電器有限公司（移動通訊產品製造商）的專家及設備，共同合作以致力發展第4代寬頻移動通訊技術在基站方面的應用。我們將集中地研發寬頻天線組件，研發功率放大器的高綫性技術，及研製嶄新的上變頻、下變頻轉化技術，以配合相控源集成電路，集成上述技術以開發新一代4G移動通訊的智能天線陣。

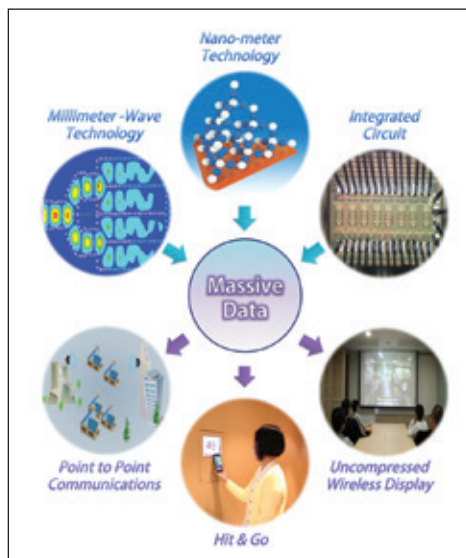
60 GHz RFIC TRANSCEIVER FOR SHORT RANGE INSTANT MASSIVE DATA SHARING

用於供線短距離實時海量數據傳輸的60 GHz 射頻集成收發前端

Industry leaders worldwide are driving WiGig/802.11ad technology standards for utilizing the 60 GHz unlicensed frequency band. Its 7 GHz absolute bandwidth allows instant massive data transmission such as uncompressed high-quality multimedia streaming, as well as sharing and backing up contents. This is the next frontier in consumer electronic connectivity in which one can replace the nest of wires in the living room with high-speed wireless technology. WWW.in-stat.com forecasted that WiGig/802.11ad-enabled PC devices shipments will approach 15 million units in 2015 and will continue to grow rapidly.

This multi-gigabit wireless system requires wideband off-chip antenna and high performance CMOS-based RF transceiver and their integration. Innovations are required to drive down the cost and enhance system performance. Integration between high-speed baseband with the RF transceiver is another essential piece to complete the jigsaw puzzle for a competitive product.

In this project, we draw upon the expertise and facilities of the SKLMW in antenna and RFIC design, together with the sales and marketing experience of Tongyu (Hong Kong) Company Limited and Zhuhai Hansen Technology Co., Ltd. to pursue the endeavor of establishing enabling technology in short range communication for high speed, high data rate communications.



目前，世界範圍內的工業界領們正在推動WiGig/802.11a技術標準的應用。這個標準利用無需牌照的6GHz頻段。該頻段具有7GHz的連續頻寬，使即時無線傳送海量資料成為可能。例如無壓縮高品質的多媒體流的傳送、備份、分享等等，都需要用到這項技術。它也均將成為下一個消費電子的熱點技術。根據WWW.in-stat.co的預測，到2015年，裝備了WiGig/802.11a的個人電腦裝置將達到一千五百萬台，而且將會持續快速增長。

這項高容量的無線系統要求寬頻晶片外置天線以及基於CMOS的射頻集成收發模組，並需要將兩者有機集成在一起。為了在這個頻實現低成本高性能的系統，我們必須開發具有原創性的天線和電路。同時，還需要解決高速基帶系統與射頻收發前端的集成問題。

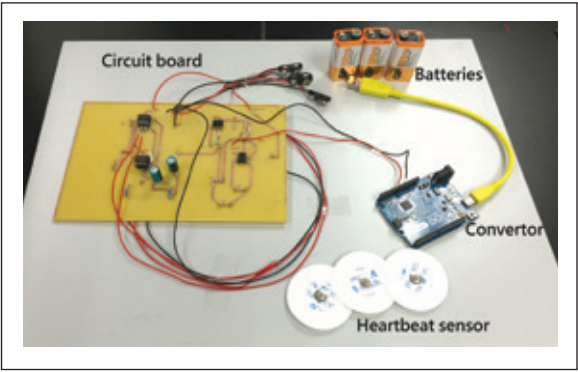
在本項目中，我們將充分利用香港城市大學毫米波國家重點實驗室多年積累的試驗設備，以及在天線和射頻積體電路方面的研究成果和經驗，開發6GHz頻段的射頻前端。我們的合作夥伴，通宇（香港）有限公司和珠海漢勝科技股份有限公司在產業化和市場推廣方面有多年的成功經驗。我們三方將共同合作，將60GHz頻段的射頻前端研發成果產業化，應用於短距離高速海量資料傳輸。

DRIVER DROWSINESS DETECTION USING HEARTBEAT

透過心跳作司機疲勞檢測

Each year nearly 1.3 million people were killed and 50 million people were injured in road traffic collision which imposed heavy loadings to healthcare industries and hospitals. It is estimated that 1% to 3% of the respective GDP of the world (more than \$500 billion) has been consumed due to traffic crashes.

This project provides an algorithm implemented into microcontroller which captures and analyzes Heartbeat signal continuously, to provide timely response and decision once driver is concluded as drowsiness.



每年大約有一百三十萬人死於及五千萬人在交通意外中受傷，這將為醫療產業及醫院造成巨大的負擔。每年全球生產總值的百份之一至三（多於5千億）用於交通意外。本項目提出了一個可安裝於微控制器的演算法，透過實時監控及分析駕駛員的心跳，演算法可即時判斷駕駛者是否有睡意。

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